Malhar Gudekar

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Education

University of Illinois Urbana Champaign August 2024 - May 2026
Master of Science, Information Management GPA: 3.8/4
• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai June 2019 - May 2023
Bachelor of Engineering, Computer Science GPA: 9.25/10
• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

Skills

- **Languages:** Python, SQL, Java(leetcode), C++(leetcode)
- **Data Engineering:** ETL/ELT Pipelines, Data Modeling, Snowflake, PySpark, Workflow Orchestration (Airflow, Step Functions, AWS Glue)
- **Cloud & Tools:** AWS (S3, EC2, EKS), PostgreSQL, Neo4j, Git, CI/CD, Docker

Work Experience

University of Illinois Urbana Champaign May 2025 - Present
Research Assistant - iSchool Champaign, USA
- Built and maintained **AWS-based scalable data pipelines** to ingest high-volume wearable data for 100+ users, enabling distributed storage in **S3** and downstream analytics consumption
- Designed and optimized **SQL-based data models and warehouse tables** in PostgreSQL, applying indexing and query tuning techniques to improve performance by 38% for large datasets
- Implemented workflow orchestration using AWS Step Functions for streaming pipelines, adding automated alerting, data quality checks, logging, error handling, and retries to ensure reliability

Research Assistant - CHI AI in Infodemic Management January 2025 - May 2025
- Designed and deployed scalable ELT pipelines in Python for high-volume datasets, implementing error handling, schema validation, and data quality monitoring, reducing latency by 41%
- Built distributed real-time ingestion pipelines using **Kafka** and **Neo4j**, enabling streaming data processing, transformation, and graph-based analytics at scale
- Applied **distributed data processing and big data principles** to handle large-scale streaming data and ensure consistent data availability across systems

Business Intelligence Group August 2025 - December 2025
Technical Consultant Champaign, USA
- Built and deployed scalable data processing pipelines with vendor API integrations using FastAPI, PostgreSQL, and optimized storage, supporting schema management and downstream analytics
- Designed fault-tolerant **ETL workflows** for ingesting and transforming high-volume document datasets, ensuring low-latency and reliable data availability
- Developed backend services and used **Git-based CI/CD workflows** to enable reliable deployment and real-time querying across distributed and high-availability environments

Swift Mobil Software Solutions Provider July 2023 - October 2023
Data Analyst Mumbai, IN
- Built **data pipelines** feeding **Power BI dashboards** for logistics KPIs, reducing reporting time by **40%** and improving data availability for business decision-making
- Processed large-scale datasets using **Python** and **PySpark distributed processing**, implementing transformations and aggregations for performance monitoring
- Developed **SQL queries** to validate, reconcile, and analyze operational data, improving data quality and speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd. January 2023 - June 2023
Data Analyst Mumbai, IN
- Built and maintained **SQL-based data pipelines** supporting **Power BI dashboards**, designing data warehouse models, writing complex joins and aggregations, and ensuring delivery of analytics-ready datasets across business functions
- Automated **data transformation workflows** using VBA, implementing validation checks, error handling, and scheduling logic to streamline large-scale data processing and cut reporting time by **42%**

Stu/Dio at Illinois August 2025 - Present
Project Manager Champaign, USA
- Led cross-functional team of **6** engineers to deliver data-driven improvements, analytics insights, and system tracking
- Managed **Agile workflows** using JIRA, tracking delivery, QA issues, and sprint progress to ensure timely and reliable releases

Projects

Scalable ML System for Customer Risk Prediction March 2025
UIUC Hackathon
- Built an **end-to-end data pipeline** using Python, Pandas, and LightGBM to process large-scale **time-series** transaction data, including ingestion, transformation, feature engineering, and model integration
- Designed and implemented **scalable ETL and feature engineering pipelines** for large datasets, capturing behavioral trends and temporal patterns to improve model performance
- Developed a **data-driven decision framework**, incorporating pipeline outputs and risk thresholds to improve prediction precision by 20% and enable scalable analytics workflows